

Report on the Workshop "Polyhedra in Lean"

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1 Stub section

Here's the hyperplane definition of a polytope over the reals, typeset using the `amsmath` package.

$$P = \{x \in \mathbb{R}^n \mid Ax \leq b\}, \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m,$$

Here's the point definition of a set being a polytope over a strictly ordered semiring in `lean[1]`.

```
variable [Semiring R] [PartialOrder R] [IsStrictOrderedRing R]
variable [ConvexSpace R X]

variable (R) in
/-- A set is a polytope if it is the convex hull of finitely
    many points. -/
def IsPolytope (s : Set X) : Prop := ∃ t : Finset X, s =
    convexHull R t
```

References

- [1] Leonardo de Moura and Sebastian Ullrich. The lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28: 28th International Conference on Automated Deduction, Virtual Event, July 12–15, 2021, Proceedings*, page 625–635, Berlin, Heidelberg, 2021. Springer-Verlag.